AUTOMATED RISK SCORING SYSTEM FOR FINANCIAL INSTITUTIONS

Natalja Krjuckova

Transport and Telecommunication Institute (TSI)
Lauvas Street 2, Riga, LV-1019, Latvia, natalja.krjuckova@gmail.com

Keywords: Synthetic Data, Risk Scoring, Fraud Detection, XAI, AML, Financial Institutions

The financial sector encounters difficulties in overseeing and identifying fraudulent operations. The Basel AML/CFT Index 2024 identifies fraud as a principal offence within banks and financial institutions (Basel AML Index, 2024). The scale of money laundering remains significant, amounting to 2-5% of global GDP annually, which is between €715 billion and €1.87 trillion per year (United Nations., 2024.). To encourage financial stability and enhance risk management, new rules and more stringent requirements for the ML risk assessment process are frequently implemented (FATF, 2025). This study aims to create an automated risk-scoring system for client transactions that will enhance financial institutions' ability to identify money laundering situations and comply with regulatory standards.

The decision to initiate the development of an automated risk assessment system involved the generation of synthetic data for the following reasons: insufficient access to extensive real-world data, coupled with the absence of data encompassing all potential suspicious circumstances, necessitates the use of synthetic data to enhance the diversity and resilience of deep learning model training (Altman *et al.*, 2023). In this context, the creation of synthetic data was used. An initial dataset of 60 distinct fraud types with a ratio of 1-5% suspicious transactions was created to simulate actual fraud trends.

Due to the limitations of real and synthetic data, several deep learning models have been applied, including Logistic regression, Random forests, and Gradient boost models, to identify the algorithm that is more accurate and efficient in detecting suspicious activity. To meet regulatory requirements and to increase model interpretability, explainable artificial intelligence (XAI) techniques, such as SHapley Additive Explanations, were employed to explain the fraud detection results (Kute *et al.*, 2021).

The implementation of such a system is also subject to various limitations; we cannot claim that this system will be unique and suitable for any type of financial institution; additionally, the following factors must be considered when implementing this system: geographic location, main business sectors, specific types of transactions, and the current percentage of true matches for suspicious transactions. As a result, financial institutions should prioritize the scenario groups and configure the model based on the above parameters, thereby the system will meet the risk profiles and regulatory requirements.

The proposed system of automated risk analysis is expected to increase the ability of financial institutions in risk identification and management, improving compliance and fraud prevention decision-making.

The research is supervised by Dr.Sc.Eng. Nadezda Spiridovska.

References

1. Basel AML Index. (2024) Basel Institute on Governance [online]. Available from: https://baselgovernance.org/blog/basel-aml-index-updates-methodology-reflect-rising-global-fraud-risks [Accessed 19 March 2025].
2. United Nations. (2024) United Nations : Office on Drugs and Crime [online]. Available from: //www.unodc.org/unodc/en/money-laundering/overview.html [Accessed 19 March 2025b].

3. FATF, (2025) *FATF Recommendations* [online]. Available from: https://www.fatf-gafi.org/en/topics/fatf-recommendations.html [Accessed 16 March 2025].
4. Altman, E., Blanuša, J., von Niederhäusern, L., Egressy, B., Anghel, A. and Atasu, K. (2023) Realistic Synthetic Financial Transactions for Anti-Money Laundering Models [online]Advances in Neural Information Processing Systems. 36. Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-85184980711&partnerID=40&md5=0ccfc15090755530ff4a6d7fe57a0154.
5. Kute, D.V., Pradhan, B., Shukla, N. and Alamri, A. (2021) Deep Learning and Explainable Artificial Intelligence Techniques Applied for Detecting Money Laundering-A Critical Review. IEEE Access [online]. 9, pp. 82300–82317. Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-85107332767&doi=10.1109%2fACCESS.2021.3086230&partnerID=40&md5=b5e30d6539fe1967b6af0d6f99c5d31a.